

BÀI TẬP Python basic 1.1

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1.input and ouput

```
In [ ]: name = input("Enter your name: ")
        print("Hello" + name + "!")
```

Hello, a!

2.Data type

```
In [2]: str = "String"
        print(str)
        x = 36
        print(x)
        y = 36.0
        print(y)
        is_valid = True
        print(is_valid)
```

String

36

36.0

True

3.DataType Advanced

```
In [ ]: # LIST
        # List Ordered, Changeble, Allow Duplicates
        list = ["banana",1,True,"banana" ]
        # first index
        print(list[0])
        # print full list
        print(list)

        # TUPLE
        # Tuple Ordered, Unchangeble, Allow Duplucate
        # You can modify tuple if you convert it to list and back original
        tuple = ("apple",2,False,"apple")
        # first index
        print(tuple[0])
        # print full tuple
        print(tuple)

        # SET
        # Set Unodered, Unchangeble, Unindexed, UnDuplicate
        set = {"baba",2,True,"baba"}
```

```

# it automatically remove items duplicate
print(set)
# Dictionary
# Dictionary Ordered, Changeable, No Duplicate
# It same like as as mapHash in java and map in GO
dict= {
    "name" : 'dvid',
    "age": 20,
}
print(dict["age"])
print(dict)

```

```

banana
['banana', 1, True, 'banana']
apple
('apple', 2, False, 'apple')
{True, 'baba', 2}
20
{'name': 'dvid', 'age': 20}

```

4.If and Loop(while and for)

```

In [5]: #If Else
age = input("Enter your age: ")
if int(age) >= 18:
    print("You are an adult.")
else:
    print("You are a minor.")

# Loop
# While Loop
# Can use countinue so that skipp the current iterationm
count = 0
while count <= 10:
    count += 2
    if count == 6:
        continue
    if count == 8:
        break
    print ("Count ", count )
else:
    print("End Count")

# For Loop
for i in "abc":
    print(i)

```

```

You are a minor.
Count 2
Count 4
a
b
c

```

5.Function

```
In [10]: #Function
def HelloWorld():
    print("This is a frist programing by myself")
HelloWord()

#Function with parameter and agurement
def Sum(x,y):
    print("This is sum ", x,"+",y,"=",x+y);

Sum(10,10)
```

This is a frist programing by myself

This is sum 10 + 10 = 20

Read and write file

```
In [ ]: #Open and read the file
```

```
In [ ]: f = open("filename.txt", "r")
content = f.read()
print(content)

# create and write to a file
with open("text1.txt","w") as f:
    f.write("Add line to the file ")
# with "w" will create a file if it does not exist

f = open("text2.txt","x")
# with "x" will create, return error if file exist

f = open("text2.txt","a")
# with "a" will append to the file if it exists

f.close()
```